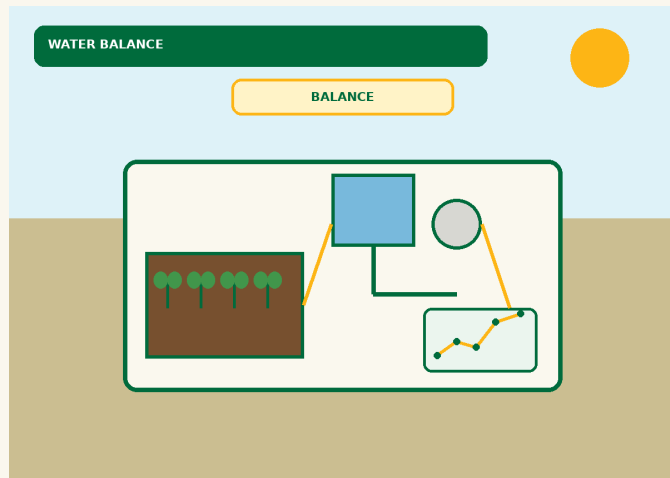




BALANCE

WATER BALANCE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

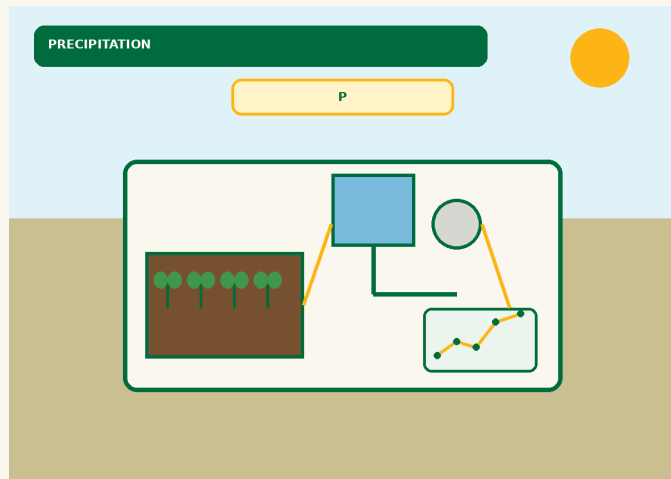
TECHNICAL CONCEPT

A water balance accounts for water inputs, outputs, and storage changes within a defined spatial and temporal boundary.

ANALYSIS / DESIGN TASK

Write the terms in a root-zone water balance.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



P

PRECIPITATION

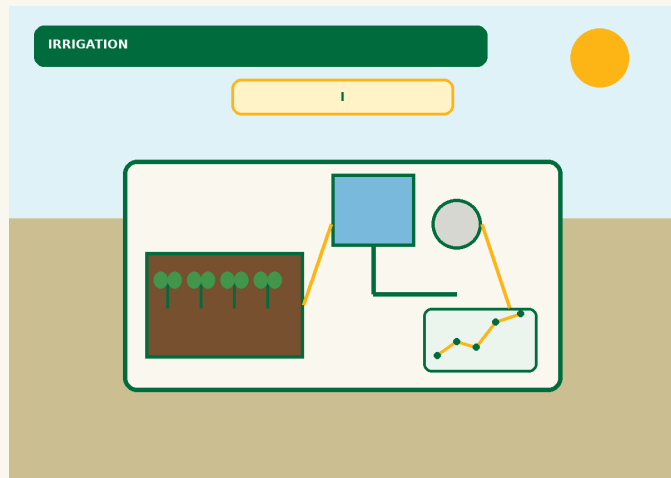
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Precipitation is atmospheric water reaching the surface and may contribute to crop water supply depending on timing and effectiveness.

ANALYSIS / DESIGN TASK

Record or interpret a precipitation amount.



I

IRRIGATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

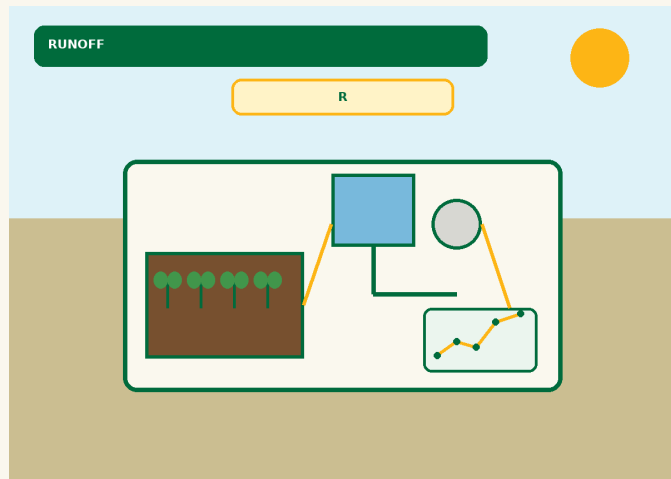
TECHNICAL CONCEPT

Irrigation is managed water applied to supplement crop water supply.

ANALYSIS / DESIGN TASK

Quantify irrigation input over a defined period.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



R

RUNOFF

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

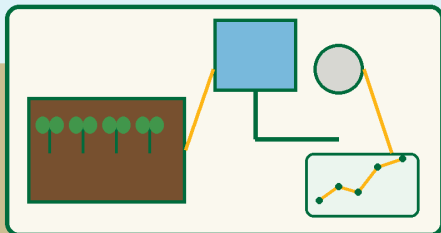
Runoff is water leaving the defined surface boundary rather than infiltrating into the root zone.

ANALYSIS / DESIGN TASK

Identify conditions increasing runoff.

DEEP PERCOLATION

DP



DP

DEEP PERCOLATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

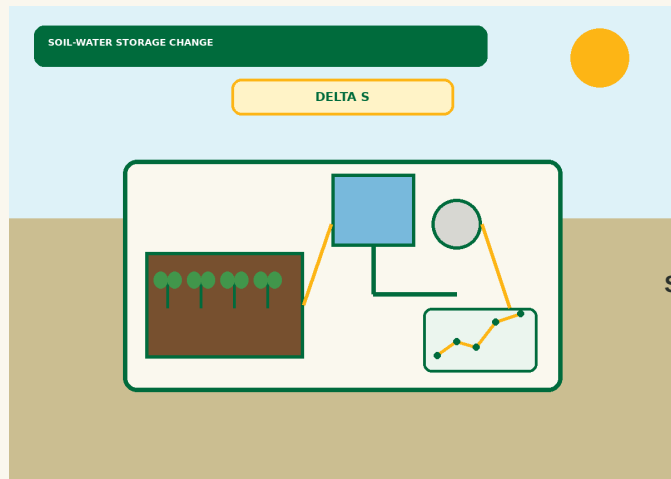
TECHNICAL CONCEPT

Deep percolation is water moving below the effective root zone and may represent drainage loss from the crop-water balance.

ANALYSIS / DESIGN TASK

Estimate it as a residual in a simplified balance.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DELTA S

SOIL-WATER STORAGE CHANGE

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Storage change is the increase or decrease in water held within the defined soil profile over a time interval.

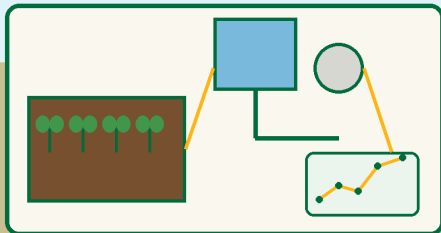
ANALYSIS / DESIGN TASK

Calculate storage change from two measurements.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

EVAPOTRANSPIRATION

ET



ET

EVAPOTRANSPIRATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

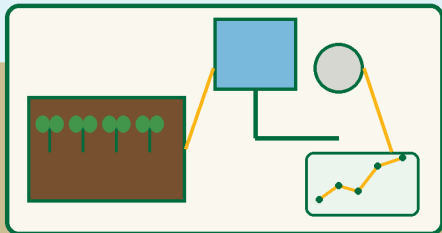
Evapotranspiration is combined evaporation and plant transpiration and is a major water-output term in cropped systems.

ANALYSIS / DESIGN TASK

Estimate ET as a residual from simplified data.

REFERENCE ET

ET₀



ET₀

REFERENCE ET

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Reference evapotranspiration is the atmospheric evaporative demand from a standardized reference surface calculated from weather variables.

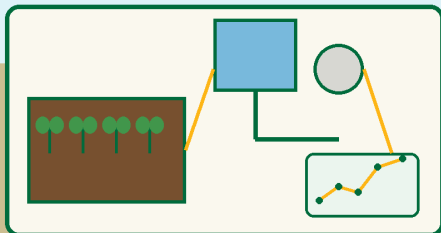
ANALYSIS / DESIGN TASK

Interpret a provided ET0 time series.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

CROP COEFFICIENT

K_c



K_c

CROP COEFFICIENT

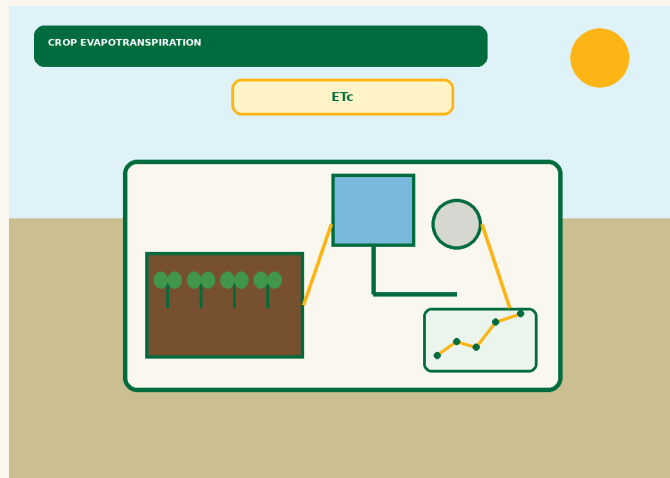
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A crop coefficient scales reference evapotranspiration to approximate crop evapotranspiration for a specified crop and growth stage.

ANALYSIS / DESIGN TASK

Calculate ET_c from ET_0 and K_c .



ETc

CROP EVAPOTRANSPIRATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Crop evapotranspiration is commonly estimated as reference evapotranspiration multiplied by an appropriate crop coefficient under stated assumptions.

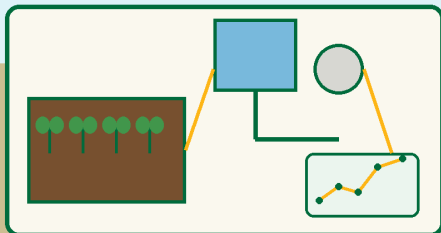
ANALYSIS / DESIGN TASK

Calculate daily ET_c .

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

IRRIGATION REQUIREMENT

IRR REQ



IRR REQ

IRRIGATION REQUIREMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Irrigation requirement is supplemental water needed after accounting for crop demand, effective water supply, and system considerations.

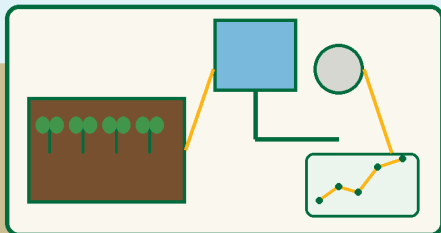
ANALYSIS / DESIGN TASK

Calculate a simplified requirement.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

EFFECTIVE WATER SUPPLY

EFFECTIVE



EFFECTIVE

EFFECTIVE WATER SUPPLY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

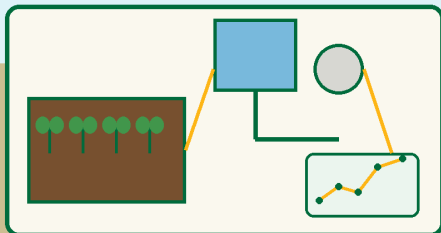
Not all supplied or received water contributes usefully to root-zone crop demand; effectiveness depends on timing, storage, runoff, and drainage.

ANALYSIS / DESIGN TASK

Compare gross and effective supply.

WATER-USE EFFICIENCY

WUE



WUE

WATER-USE EFFICIENCY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

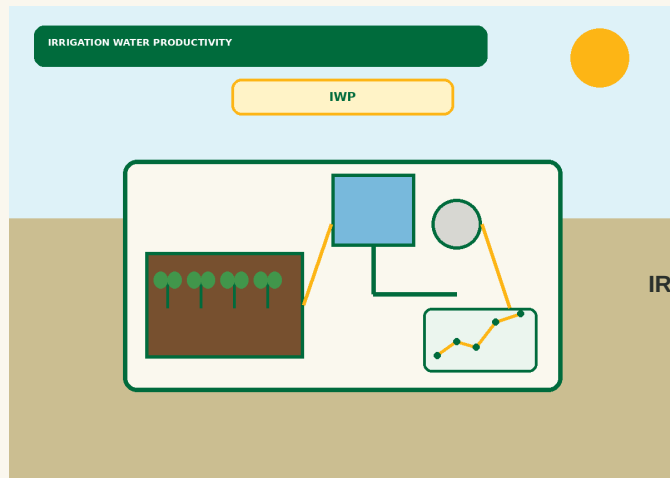
TECHNICAL CONCEPT

Water-use efficiency is a family of metrics relating biomass or yield to water use; the denominator must be explicitly defined.

ANALYSIS / DESIGN TASK

Compare two WUE calculations with different denominators.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



IWP

IRRIGATION WATER PRODUCTIVITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

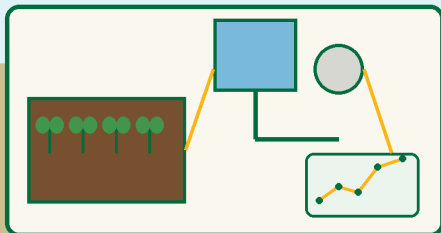
Irrigation water productivity relates crop output specifically to irrigation water supplied.

ANALYSIS / DESIGN TASK

Calculate kilograms per cubic metre of irrigation.

ET WATER PRODUCTIVITY

ETWP



ETWP

ET WATER PRODUCTIVITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

ET water productivity relates crop output to evapotranspiration and differs conceptually from irrigation-based productivity.

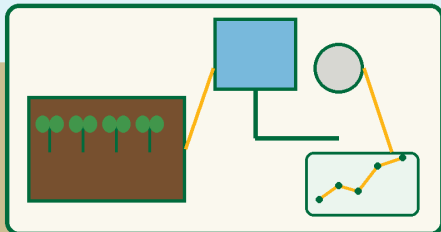
ANALYSIS / DESIGN TASK

Calculate and compare ETWP.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

MARGINAL WATER PRODUCTIVITY

MARGINAL WP



MARGINAL WP

MARGINAL WATER PRODUCTIVITY

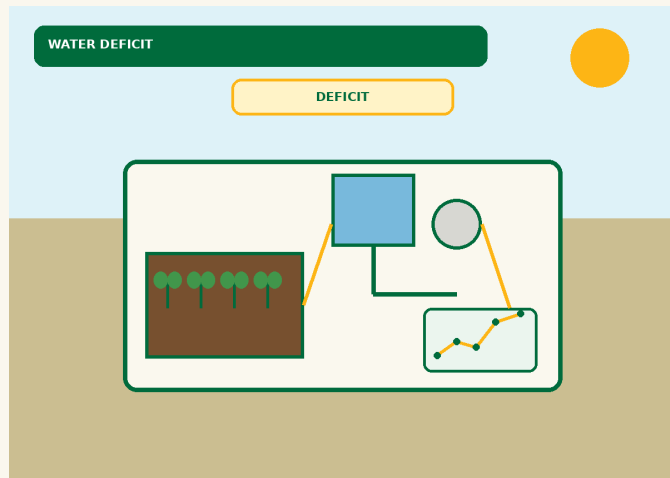
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Marginal water productivity describes additional output associated with an additional unit of water over a specified comparison.

ANALYSIS / DESIGN TASK

Calculate from two treatment levels.



DEFICIT

WATER DEFICIT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

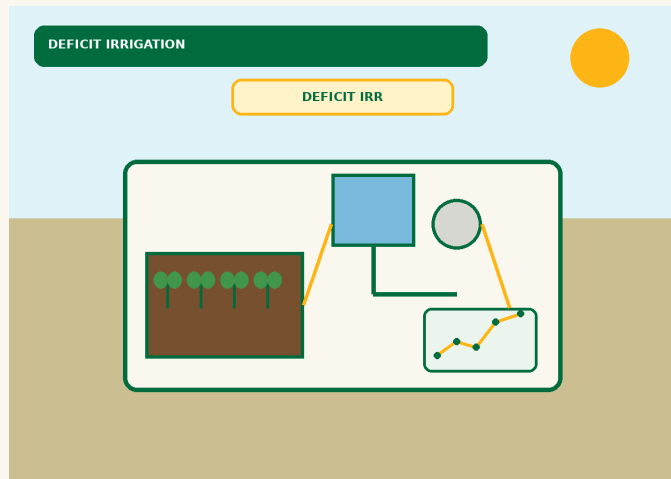
TECHNICAL CONCEPT

Water deficit occurs when crop water supply is below demand and may reduce growth or yield depending on timing and severity.

ANALYSIS / DESIGN TASK

Interpret a deficit scenario.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



DEFICIT IRR

DEFICIT IRRIGATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Deficit irrigation deliberately supplies less than full estimated crop water requirement to achieve defined production or resource objectives.

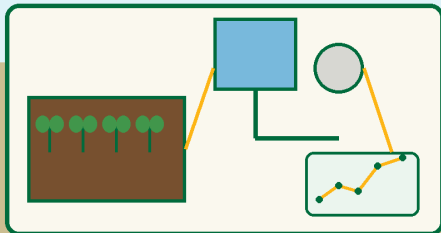
ANALYSIS / DESIGN TASK

Evaluate a simplified deficit strategy.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

YIELD RESPONSE TO WATER

RESPONSE



RESPONSE

YIELD RESPONSE TO WATER

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

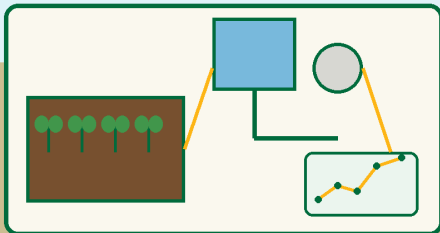
Yield response to water describes how crop yield changes with water availability and may differ across growth stages.

ANALYSIS / DESIGN TASK

Interpret a response curve.

WATER PRODUCTIVITY OPTIMIZATION

OPTIMIZE



OPTIMIZE

WATER PRODUCTIVITY OPTIMIZATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Optimization seeks a water-management strategy that improves defined production and resource objectives under supply and system constraints.

ANALYSIS / DESIGN TASK

Recommend a strategy using water and yield data.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE